

Smart Internship Management System

Database Systems Lab — Project Documentation

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Version 2.0 — 16 May 2026

Milestone Tracking:

Milestones	Date	Version	Remarks
ERD Diagram and Schema (1.1 — Updated)	24/4/2026	1.1	
ERD Diagram and Schema (Updated)	10/5/2026	1.1	
Applied 1NF,2NF and 3NF normalization and updated ERD	16/5/2026	2.0	
Dataflow Description	16/5/2026	3.0	

Milestone 3 — Dataflow Description

This section describes exactly where data enters the Smart Internship Management System, how it flows through the database tables, and what outputs the system produces.

1. Where Data Enters the System

Student Registration

A student fills in their profile: Name, Email, Phone, Department, CGPA, Semester, DegreeProgram.

- Data stored in: Students table. Student's selected skills are stored in: StudentSkills table (linking Students to Skills).

Company Registration

A company provides: CompanyName, Location, ContactEmail, IndustryType, Website, HRContact.

- Data stored in: Companies table.

Internship Posting

A company posts an internship with Title, Description, Deadline, InternshipType, Duration, Stipend, WorkMode, InternshipCategory.

- Data stored in: Internships table (with CompanyID as foreign key linking to the posting company).

Eligibility Setup

Admin sets MinCGPA, MinSemester, ExperienceLevel for each internship, selects required skills, and specifies which departments may apply.

- Eligibility rules stored in: EligibilityCriteria table (1:1 with Internships).
- Required skills stored in: InternshipSkills table (linking Internships to Skills).
- Eligible departments stored in: EligibleDepartments table (linking Internships to Departments).

2. How Data Moves Through the System

When a Student Applies for an Internship

- System reads Students table — retrieves student CGPA, Semester, and Department.
- System reads EligibilityCriteria — checks if student CGPA $\geq$ MinCGPA and Semester $\geq$ MinSemester.
- System reads EligibleDepartments joined with Departments — checks if student's department is in the allowed list.
- If all checks pass — a new row is inserted into Applications with Status = Pending.
- If any check fails — Status is automatically set to Ineligible, preventing the application from reaching the company.

When a Company Reviews Applications

- Applications table is joined with Students table — to display applicant name, CGPA, and department.
- Applications table is joined with Internships table — to confirm which role the student applied for.
- Company updates the Application Status to Accepted or Rejected in the Applications table.

When Feedback is Submitted

- Student selects the Company and Internship they completed.
- Rating and Comments are inserted into the Feedback table with StudentID, CompanyID, and InternshipID as foreign keys.

3. What Comes Out of the System

- Eligible Applicants List — generated by joining Applications, Students, and EligibilityCriteria for a given internship.
- Application Status Report — a student can view all their applications and current statuses from the Applications table.
- Company Feedback Summary — average rating per company queried from the Feedback table grouped by CompanyID.
- Skills Match Report — generated by comparing StudentSkills with InternshipSkills to show how well a student matches an internship.
- Admin Dashboard — shows all pending applications by querying Applications where Status = Pending, joined with student and internship details.